DEXAMETHASONE

No mineralocorticoid effect

Fetal lung maturity

CHEMICAL NAME 9-fluoro-11β,17,21-trihydroxy-16α-methylpregna-1,4-diene-3,20-dione (sodium phosphate)

CLASS Long-acting potent corticosteroid (glucocorticoid)

TRADE NAMES Decadron · Dexa · Fortecortin · Maxidex (eye)

THERAPEUTIC USES / INDICATIONS

- Cerebral oedema (tumour), raised ICP.
- Croup (single oral dose); bacterial meningitis (before/with first antibiotic dose).
- Antenatal fetal lung maturation (preterm labour 24–34 weeks).
- Chemotherapy-induced nausea and vomiting; PONV prophylaxis; severe COVID-19 requiring oxygen; airway oedema; spinal cord compression.
- Anaphylaxis/allergy adjunct; Cushing's diagnosis (suppression test).

ADMINISTRATION

- **IV** slowly (rapid → perineal burning/itching), **IM**, **PO** with food.
- Eye drops.
- Note dose units: dexamethasone vs dexamethasone phosphate (3.3 mg base = 4 mg phosphate).

DOSAGES

- **Croup:** 0.15 mg/kg PO single dose (up to 0.6 mg/kg severe).
- **Antenatal (preterm labour):** 6 mg IM every 12 h × 4 doses (or betamethasone).
- **Cerebral oedema:** 10 mg IV, then 4 mg every 6 h.
- **Meningitis:** 0.15 mg/kg IV every 6 h for 4 days (max 10 mg).
- **PONV:** 4–8 mg IV at induction; **COVID-19:** 6 mg daily for 10 days.

PHARMACOKINETICS

- **Onset:** IV minutes; PO 1–2 h. **Duration:** 36–72 h (long-acting).
- Metabolised in liver (CYP3A4).
- Excreted in urine. **t½:** 3–4 h (biological 36–54 h).

PHARMACODYNAMICS

- Potent **glucocorticoid** (~25–30× hydrocortisone) with no mineralocorticoid effect → minimal fluid retention.
- Crosses placenta (not inactivated) → stimulates surfactant production in fetal lungs.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Hyperglycaemia, insomnia, agitation.
- Increased appetite, dyspepsia.
- Perineal burning with rapid IV.

ADVERSE EFFECTS (SERIOUS)

- Adrenal suppression (long-acting; significant).
- Psychosis; infection; GI bleeding.
- Myopathy, osteoporosis (long-term).
- Neonatal hypoglycaemia after antenatal doses.

CONTRAINDICATIONS & CAUTIONS

- Systemic infection without cover (except meningitis protocol).
- Live vaccines at immunosuppressive doses.
- **Caution:** diabetes, peptic ulcer, psychiatric disorders.

KEY INTERACTIONS

- **Enzyme inducers (phenytoin, rifampicin):** ↓ levels.
- **Antidiabetics:** ↑ glucose.
- **Anticoagulants:** altered INR.
- **NSAIDs:** GI bleeding.

NURSING MANAGEMENT

- Monitor glucose (even in non-diabetics), especially in pregnancy and cancer.
- Antenatal: record time of doses; inform neonatal team; monitor maternal glucose.
- IV slowly to avoid perineal discomfort.
- Croup: reassess stridor and work of breathing.

PATIENT EDUCATION

- Take with food in the morning.
- Report high glucose, mood changes, signs of infection.
- Do not stop long courses abruptly.

KEY POINTS

- Potent long-acting glucocorticoid, no mineralocorticoid effect.
- Cerebral oedema, meningitis, PONV.
- Croup 0.15 mg/kg once; antenatal 6 mg IM × 4.
- Hyperglycaemia.